



COURSE SLIDES

Project Management Professional (PMP) Exam Prep

Course Code: C523

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COURSE ADMINISTRATION

Welcome & Housekeeping

About the Trainer



Your Trainer

General Trainer template —
to be completed by the trainer

NAME

TITLE / DESIGNATION

QUALIFICATIONS

AREAS OF EXPERTISE

TRAINING & INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

ROLE

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

CERTIFICATION

Industry certified in the subject area of this course.

DELIVERS

Professional short courses for Tertiary Infotech Academy.

FOUNDER

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

Let's Know Each Other

1 Your name and organisation / role.

2 Your experience with this subject (if any).

3 What you want to be able to do after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

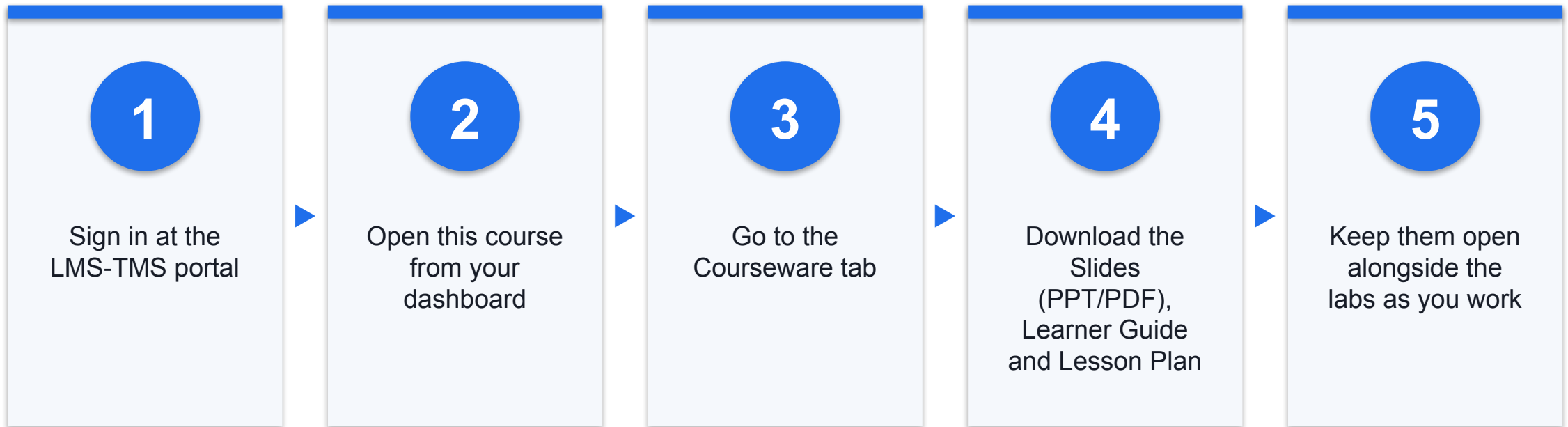
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Download Course Material



Portal: <https://lms-tms.tertiaryinfotech.com>

Lesson Plan — 4 Days, 8 hours/day

Day 1

- Day 1 — Business Environment & Starting the Project
 - Topic 1: Business Environment (Labs 1–4)
 - Topic 2: Start the Project (Labs 5–8)

Day 2

- Day 2 — Planning the Project
 - Topic 3: Plan the Project (Labs 9–14)
- Daily timing
 - 9:30am–6:30pm · 1-hour lunch · tea breaks within training time

Learning Outcomes

1

Scope & Authorise

Scope project requirements and build a business case, charter and traceable scope baseline that authorise delivery.

2

Plan & Baseline

Develop an integrated project plan with realistic schedule, cost and resource baselines using PERT, critical path and reserve analysis.

3

Lead the Team

Lead a project team through its development stages, applying leadership style, motivation theory and structured conflict resolution.

4

Risk & Stakeholders

Analyse project risk quantitatively and engage stakeholders through a planned communication and escalation cadence.

5

Control & Close

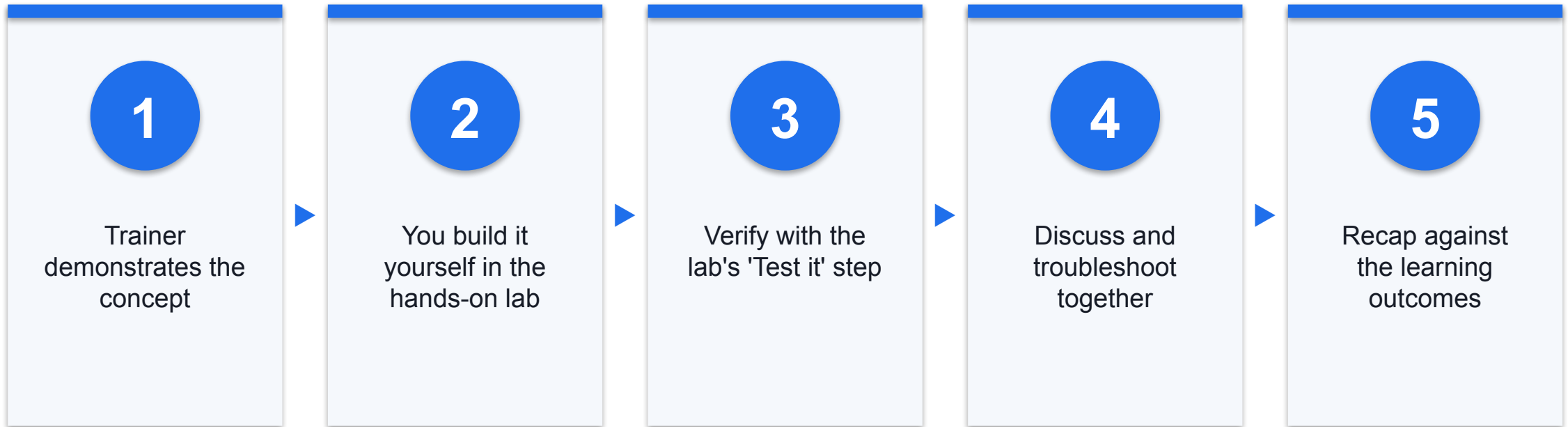
Control delivery against baselines with earned value, flow metrics and statistical process control, then close the project and realise benefits.

6

Exam Readiness

Apply the PMI PMP Examination Content Outline (ECO 2026) across all three domains and sit a full-format mock exam with domain score analysis.

How You'll Learn





CORE CONCEPTS

Course Fundamentals

The PMP Exam at a Glance

1

180 questions, 230 minutes

170 scored plus 10 unscored pretest items, with two 10-minute breaks.

2

Three domains

People 33%, Process 41%, Business Environment 26% — weighted by the ECO 2026.

3

Predictive, agile and hybrid

About 40% predictive; the other 60% splits between adaptive/agile and hybrid.

4

Six question formats

Multiple choice and response, matching, enhanced matching, graphic-based and case study.

5

Scenario-driven, not recall

Almost every item is a situation asking what you do next, not a definition.

6

35 contact hours

This course delivers the 35 hours of project management education PMI requires.

The ECO 2026 Domains

1

People — 33%

Lead, align and communicate with the humans on and around the project.

2

Process — 41%

Plan, deliver, measure and close the work itself.

3

Business Environment — 26%

Connect the project to governance, compliance, risk and change.

HOW THIS COURSE WORKS

The exam does not ask what you know. It asks what you would do.

Every lab in this course puts you in a scenario, makes you choose, and then shows you why the exam would score that choice the way it does.

What You'll Build

An authorised project

- Environment scan and governance model
- Compliance register
- Business case with NPV and BCR
- Project charter and stakeholder analysis

An integrated plan

- Requirements traceability matrix
- Prioritised backlog and WBS
- Critical path with PERT estimates
- Cost baseline with reserves and a risk register

A controlled delivery

- RACI and communication plan
- Kanban flow metrics
- Earned value analysis and forecasts
- Root cause analysis and control charts

How Every Lab Progresses

1

Read the scenario and identify which development approach is in play.

2

Work the technique by hand — the exam gives you no spreadsheet.

3

Produce the artifact the next lab will consume.

4

Verify your result against the stated success test.

5

Connect it to the exam: what would the item stem look like, and what is the tell?

TOPIC 01

Business Environment

Foundations · governance · compliance · change · development approaches

Key Concepts — Business Environment

1

A project is a temporary endeavour creating a unique product, service or result — it is bounded in time and delivers change, unlike operations.

2

The 12 PMI principles and the triple constraint frame every tailoring decision you make on the exam and on the job.

3

Two life cycles, five approaches: the project life cycle is the phases; the development life cycle is predictive, iterative, incremental, agile or hybrid.

4

EEFs constrain you and OPAs are reusable assets — you conform to EEFs, you apply and then update OPAs.

5

Governance sets escalation thresholds: escalate outside tolerance with options, resolve inside tolerance with the team; phase gates decide go, change, hold or kill.

6

Compliance, sustainability and the responsible use of AI are the project manager's own obligations, not someone else's department.

Hands-On Labs — Business Environment

Labs 1–2

- PMP Exam Orientation and 35 PDU Study Plan
- PESTLE and TECOP External Environment Scan

Labs 3–4

- EEF/OPA Inventory, Governance and Escalation Thresholds
- Compliance, Sustainability and AI-Governance Register

PMP Exam Orientation and 35 PDU Study Plan

LAB 1

Reconstruct the ECO (July 2026) blueprint from its totals - 180 questions, 170 scored, 240 minutes - deriving the per-domain scored counts and the 80-second pacing rule by hand. Catalogue all six question types including graphic-based items, record the 35 contact-hour evidence, and build a six-week study plan weighted to the domain percentages. Test answer-selection heuristics on three Contoso case scenarios.

You'll build

An exam blueprint with derived question counts and pacing figure (artifacts/01-exam-blueprint.md), a 35 contact-hour record (artifacts/01-pdu-log.md), a six-week weighted study plan (artifacts/01-study-plan.md), answer-selection heuristics and a baseline score.

Tools: Spreadsheet, Markdown editor

PMP Exam Orientation and 35 PDU Study Plan

STEP 1 OF 7



1

Build the exam blueprint from the ECO 2026 numbers and derive the per-domain scored-question counts

PMP Exam Orientation and 35 PDU Study Plan

STEP 2 OF 7



2

Log the six question types and write an approach for each under time pressure

PMP Exam Orientation and 35 PDU Study Plan

STEP 3 OF 7



3

Record your 35 contact-hour evidence against the PMP eligibility rule

PMP Exam Orientation and 35 PDU Study Plan

STEP 4 OF 7



4

Build a six-week study plan weighted to the blueprint domain percentages

PMP Exam Orientation and 35 PDU Study Plan

STEP 5 OF 7



5

Write your answer-selection heuristics: what to eliminate and what to prefer

PMP Exam Orientation and 35 PDU Study Plan

STEP 6 OF 7



6

Apply the heuristics to three scenarios, naming the heuristic that decided each

PMP Exam Orientation and 35 PDU Study Plan

STEP 7 OF 7



Set your baseline score over 20 timed practice questions

PMP Exam Orientation and 35 PDU Study Plan

Test it

The per-domain counts read 56 / 70 / 44 and sum to exactly 170; the pacing figure is 80 seconds per question derived from 240 / 180; the study plan gives Process the largest single hour allocation and contains two full-length timed mocks in the final week; and each scenario answer names a heuristic rather than only a letter.

PESTLE and TECOP External Environment Scan

LAB 2

From the Contoso briefing pack, run a PESTLE scan across all six categories, scoring every factor for impact and likelihood. Run a TECOP scan over the five risk dimensions to surface the operational and political exposure PESTLE misses. Combine and rank both tables, take the vital few scoring 16 or above, and translate each into a scope, backlog or cost consequence with a named destination artifact.

You'll build

A scored six-category PESTLE scan (artifacts/02-pestle-scan.md), a scored five-dimension TECOP scan (artifacts/02-tecop-scan.md), a combined ranked factor table, and a backlog-impact table mapping each top factor to its destination artifact (artifacts/02-backlog-impacts.md).

Tools: PESTLE, TECOP, Mind Mapping

PESTLE and TECOP External Environment Scan

STEP 1 OF 7



1

Read the Contoso operating-environment briefing pack

PESTLE and TECOP External Environment Scan

STEP 2 OF 7



2

Build and score the PESTLE table, adding at least two factors of your own

PESTLE and TECOP External Environment Scan

STEP 3 OF 7



3

Run the TECOP scan across the technical, environmental, commercial, operational and political dimensions

PESTLE and TECOP External Environment Scan

STEP 4 OF 7



4

Rank the combined factors and take the vital few scoring 16 or above

PESTLE and TECOP External Environment Scan

STEP 5 OF 7



5

Translate each ranked factor into a scope and backlog consequence with a destination artifact

PESTLE and TECOP External Environment Scan

STEP 6 OF 7



6

Set the review cadence, owner and out-of-cycle triggers

PESTLE and TECOP External Environment Scan

STEP 7 OF 7

7

Answer the exam-style scenarios on responding to external change

PESTLE and TECOP External Environment Scan

Test it

Every PESTLE letter carries at least one factor, every score is the product of a stated impact and likelihood, the TECOP table surfaced at least one factor absent from the PESTLE table, and every factor scoring 16 or above has a named destination artifact in a later lab rather than a vague instruction to monitor.

EEF/OPA Inventory, Governance and Escalation Thresholds

LAB 3

Separate Enterprise Environmental Factors - conditions the project must live with - from Organizational Process Assets, the templates and standards it reuses. Design the Contoso governance model: which body decides what, at which cadence, with what rights. Set numeric escalation thresholds for cost, schedule, risk and issue severity, and separate the change control path from the issue resolution path.

You'll build

A classified EEF/OPA inventory (artifacts/03-eef-opa-inventory.md), a governance model naming decision bodies, rights, cadence and success metrics (artifacts/03-governance-model.md), and a numeric escalation threshold table separating change and issue paths (artifacts/03-escalation-thresholds.md).

Tools: Spreadsheet, Markdown editor

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 1 OF 7



Inventory the Contoso Enterprise Environmental Factors

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 2 OF 7

A green circle with a white number 2 inside, indicating the second step of the process.

Inventory the Contoso Organizational Process Assets

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 3 OF 7



3

Design the governance model: bodies, decision rights and cadence

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 4 OF 7



4

Define the project success metrics and their measurement baselines

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 5 OF 7



5

Set numeric escalation thresholds for cost, schedule, risk and issues

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 6 OF 7



6

Separate the change control path from the issue resolution path

EEF/OPA Inventory, Governance and Escalation Thresholds

STEP 7 OF 7

7

Answer the exam-style scenarios on governance and escalation

EEF/OPA Inventory, Governance and Escalation Thresholds

Test it

Every item in the inventory is classified as an EEF or an OPA and the learner can justify each classification; each governance body has a named decision right and cadence; every escalation threshold carries a number rather than a word such as significant; and the change path and the issue path are visibly different routes with different owners.

Compliance, Sustainability and AI-Governance Register

LAB 4

Classify Contoso's compliance requirements - PDPA personal-data protection, the statutory seven-year attendance retention obligation and the pre-go-live review - then identify the threats to each and quantify the consequences of noncompliance. Decide the actions and how compliance is measured. Add the two areas the July 2026 ECO expects: net-zero hosting, and AI governance for the deferred AI recommender.

You'll build

A compliance register with threats, consequences, actions and measures (artifacts/04-compliance-register.md), a sustainability plan covering the net-zero hosting obligation (artifacts/04-sustainability-plan.md), and an AI-governance register (artifacts/04-ai-governance.md).

Tools: Spreadsheet, Markdown editor

Compliance, Sustainability and AI-Governance Register

STEP 1 OF 8



1

Confirm the compliance requirements and classify them by category

Compliance, Sustainability and AI-Governance Register

STEP 2 OF 8



2

Identify the threats to each compliance requirement

Compliance, Sustainability and AI-Governance Register

STEP 3 OF 8



3

Analyse and quantify the consequences of noncompliance

Compliance, Sustainability and AI-Governance Register

STEP 4 OF 8



4

Decide the approach and the specific actions for each requirement

Compliance, Sustainability and AI-Governance Register

STEP 5 OF 8



5

Define how compliance is measured, with evidence and a measurement owner

Compliance, Sustainability and AI-Governance Register

STEP 6 OF 8



6

Build the sustainability section against the group net-zero obligation

Compliance, Sustainability and AI-Governance Register

STEP 7 OF 8



Build the AI-governance register for the deferred AI recommender

Compliance, Sustainability and AI-Governance Register

STEP 8 OF 8



8

Answer the exam-style scenarios on compliance decisions

Compliance, Sustainability and AI-Governance Register

Test it

Every compliance requirement has a named threat, a quantified consequence, an action and a measurement method with an owner; the sustainability section ties to a reportable hosting carbon figure; and the AI-governance register records data use, human oversight and an accountable owner rather than a general statement of intent.

Recap — Business Environment

1

You can now: Business Env T1 - Define and establish project governance (success metrics)

2

You can now: Business Env T8 - Evaluate external business environment changes; Business Env T7 - Support organizational change

3

You can now: Business Env T1 - Define and establish project governance; Business Env T4 - Remove impediments and manage issues

4

You can now: Business Env T2 - Plan and manage project compliance; Process T7 - Plan and optimize quality (regulatory compliance, sustainability)

TOPIC 02

Start the Project

Business case · charter · stakeholders · team formation

Key Concepts — Start the Project

1

The business case justifies the project; the charter authorises it. The business case can exist without a project — the charter cannot.

2

Cost-benefit analysis with NPV, payback and BCR turns a proposal into a defensible investment decision.

3

The charter is issued by the sponsor, names the project manager and states their authority level — it is high level and rarely changed.

4

Stakeholder identification, power/interest analysis and the salience model determine who gets what engagement and how often.

5

The team charter establishes ground rules and a shared vision before the first conflict, not after it.

6

The development approach is a documented tailoring decision justified by project context — never a personal preference.

Hands-On Labs — Start the Project

Labs 5–6

- Business Case and Cost-Benefit Analysis
- Project Charter

Labs 7–8

- Stakeholder Register, Power/Interest Grid and Salience Model
- Team Charter, Ground Rules and Shared Vision

Business Case and Cost-Benefit Analysis

LAB 5

Turn the Contoso baseline metrics into a quantified benefits model, then build the cost side against the board-approved SGD 480,000 ceiling. Compute payback, NPV, ROI and benefit-cost ratio by hand, including the year-zero outflow most commonly dropped from NPV. Stress-test against the Lab 02 downside of an 8% fall in enrolment, then choose among options including a costed do-nothing.

You'll build

A quantified cost-benefit model with payback, NPV, ROI and BCR plus a downside sensitivity run (artifacts/05-cba-model.md), a written business case with a recommendation and its deciding measure (artifacts/05-business-case.md), and a benefits realisation map (artifacts/05-benefits-map.md).

Tools: Statistics, Spreadsheet

Business Case and Cost-Benefit Analysis

STEP 1 OF 8



1

Quantify the benefits from the Lab 03 baseline metrics

Business Case and Cost-Benefit Analysis

STEP 2 OF 8



2

Build the cost side against the board-approved budget ceiling

Business Case and Cost-Benefit Analysis

STEP 3 OF 8

3

Compute payback period, NPV, ROI and benefit-cost ratio by hand

Business Case and Cost-Benefit Analysis

STEP 4 OF 8



4

Stress-test the case against the Lab 02 downside enrolment scenario

Business Case and Cost-Benefit Analysis

STEP 5 OF 8



5

Choose among the project options, including a costed do-nothing option

Business Case and Cost-Benefit Analysis

STEP 6 OF 8

A green circle with a white number 6 inside, indicating the current step in the process.

Write the business case document with a recommendation

Business Case and Cost-Benefit Analysis

STEP 7 OF 8



Build the benefits realisation map for later closure measurement

Business Case and Cost-Benefit Analysis

STEP 8 OF 8



8

Answer the exam-style scenarios on value and sunk cost

Business Case and Cost-Benefit Analysis

Test it

Every benefit line traces to a Lab 03 baseline metric; all discount factors fall below 1 and decrease with time; the NPV subtracts the year-zero investment; the BCR is total benefit over total cost and exceeds 1; the do-nothing option carries a negative number rather than a blank; and the recommendation cites NPV as the deciding measure.

Project Charter

LAB 6

Assemble the governance model, compliance requirements and business-case financials into a charter that authorises the project. State success criteria each carrying a numeric baseline and target, and define scope boundaries including exclusions with traceable rationales. Name the PM's procurement and change-approval authority in figures matching the Lab 03 thresholds, and open the assumption log.

You'll build

A project charter with measurable success criteria, scope boundaries, exclusions, milestones, budget and PM authority (artifacts/06-project-charter.md), a development-approach decision (artifacts/06-approach-decision.md) and an assumption log with validation dates (artifacts/06-assumption-log.md).

Tools: Markdown editor, Spreadsheet

Project Charter

STEP 1 OF 9



Assemble the charter inputs from the governance, compliance and business-case artifacts

Project Charter

STEP 2 OF 9



2

Write the charter header and the purpose statement

Project Charter

STEP 3 OF 9



3

State measurable success criteria with numeric baselines and targets

Project Charter

STEP 4 OF 9



4

Define scope boundaries, including exclusions and their rationale

Project Charter

STEP 5 OF 9



5

Record the high-level requirements, milestones and budget

Project Charter

STEP 6 OF 9



6

Name the project manager's authority in specific procurement and change-approval figures

Project Charter

STEP 7 OF 9

7

Document the hybrid development approach as a tailoring decision

Project Charter

STEP 8 OF 9



Build the assumption and constraint log with validation dates

Project Charter

STEP 9 OF 9



Answer the exam-style scenarios on charter authority and scope

Project Charter

Test it

Every success criterion carries a numeric baseline and target; the charter has a sponsor signature block; every exclusion has a traceable rationale; the milestone list contains milestones only and no durations; the approach decision names what is predictive, what is adaptive and the interface between them; and every assumption has a validation date before being wrong becomes unrecoverable.

Stakeholder Register, Power/Interest Grid and Salience Model

LAB 7

Identify the full Contoso stakeholder set, including forgotten groups such as the three admin staff who are also subject matter experts. Analyse each on power, interest, attitude and impact, plot the power/interest grid and derive the strategy per quadrant. Where the grid fails, apply the salience model of power, legitimacy and urgency. Build the current-versus-desired engagement matrix and close each gap.

You'll build

A stakeholder register scored on power, interest, attitude and impact (artifacts/07-stakeholder-register.md), a power/interest grid (artifacts/07-power-interest-grid.md), a salience analysis (artifacts/07-salience-model.md) and an engagement matrix (artifacts/07-engagement-matrix.md).

Tools: Power/Interest Grid, Salience Model, Spreadsheet

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 1 OF 8



Identify the full stakeholder set, including the groups projects habitually forget

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 2 OF 8



2

Build the stakeholder register with power, interest, attitude and impact

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 3 OF 8



3

Plot the power/interest grid and derive the strategy each quadrant demands

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 4 OF 8

4

Apply the salience model where the grid fails to give a usable answer

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 5 OF 8



5

Build the current-versus-desired engagement matrix

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 6 OF 8



6

Surface and plan to resolve the misaligned stakeholder expectations

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 7 OF 8



Set a tailored engagement cadence per stakeholder group

Stakeholder Register, Power/Interest Grid and Salience Model

STEP 8 OF 8



8

Answer the exam-style scenarios on stakeholder engagement

Stakeholder Register, Power/Interest Grid and Salience Model

Test it

The register includes the three admin staff; every stakeholder has numeric power and interest scores matching their grid position; the salience analysis produced at least one insight the grid did not; every current-to-desired gap has an action with an owner and date; the expectation-conflict plan attaches a trade-off number; and the cadence differs by stakeholder rather than one fortnightly email for everyone.

Team Charter, Ground Rules and Shared Vision

LAB 8

Draft a shared vision short enough to recall without reading it, traced to the Lab 06 success criteria. Build the team charter: name all nine members, write the working agreement on core hours and time zones, and set ground rules across six categories, each with an observable violation test. Agree a definition of done separating done from released, and set the decision rule - consent, consensus or command.

You'll build

A vision statement traced to the charter success criteria (artifacts/08-vision-statement.md), a team charter with the working agreement, definition of done, decision-rule table and skills matrix (artifacts/08-team-charter.md), and six ground-rule categories (artifacts/08-ground-rules.md).

Tools: Team Charter, Definition of Done, Skills Matrix

Team Charter, Ground Rules and Shared Vision

STEP 1 OF 10



1

Draft the shared vision statement and test it against the criteria

Team Charter, Ground Rules and Shared Vision

STEP 2 OF 10



2

Name the team and record each member's role

Team Charter, Ground Rules and Shared Vision

STEP 3 OF 10



3

Write the working agreement on core hours, time zones and availability

Team Charter, Ground Rules and Shared Vision

STEP 4 OF 10



4

Set ground rules across the six categories, each with an observable standard

Team Charter, Ground Rules and Shared Vision

STEP 5 OF 10



5

Agree the definition of done that QA and the developers will both accept

Team Charter, Ground Rules and Shared Vision

STEP 6 OF 10



6

Choose the decision-making rule for each class of decision and name the decider

Team Charter, Ground Rules and Shared Vision

STEP 7 OF 10



7

Build the skills and capacity matrix and expose the single points of failure

Team Charter, Ground Rules and Shared Vision

STEP 8 OF 10



8

Design the graduated response to ground-rule violations

Team Charter, Ground Rules and Shared Vision

STEP 9 OF 10



9

Forward-reference the responsibility assignment matrix and set the vision review cadence

Team Charter, Ground Rules and Shared Vision

STEP 10 OF 10

10

Answer the exam-style scenarios on team leadership and ground rules

Team Charter, Ground Rules and Shared Vision

Test it

The vision is under 30 words and traces to at least two numbered charter success criteria; every ground rule has a violation test; the definition of done separates done from released; the decision table uses at least two rules and names a decider for every command-rule row; and the skills matrix identifies at least three single points of failure, each with a named owner and dated mitigation.

Recap — Start the Project

1

You can now: Process T3 - Help ensure value-based delivery; Process T6 - Plan and manage finance

2

You can now: Process T1 - Develop an integrated project management plan and plan delivery; Process T2 - Develop and manage project scope

3

You can now: People T4 - Engage stakeholders; People T5 - Align stakeholder expectations; People T6 - Manage stakeholder expectations

4

You can now: People T1 - Develop a common vision; People T3 - Lead the project team

TOPIC 03

Plan the Project

Scope · schedule · cost · quality · resources · communications · risk

Key Concepts — Plan the Project

1

The integrated project management plan is the sum of subsidiary plans plus the scope, schedule and cost baselines.

2

Requirements elicitation feeds a requirements traceability matrix so every deliverable traces back to a business need.

3

MoSCoW and Kano prioritise the backlog by value; the WBS decomposes scope to work packages with a WBS dictionary.

4

PERT three-point estimating and critical path analysis expose which activities actually drive the finish date and where float exists.

5

Cost estimating produces a budget with contingency reserve for known unknowns and management reserve for unknown unknowns.

6

Risk identification, qualitative and quantitative analysis with EMV and decision trees produce a risk register with owned responses.

Hands-On Labs — Plan the Project

Labs 9–10

- Requirements Elicitation, Mind Mapping and the Traceability Matrix
- MoSCoW, Kano and the Product Backlog

Labs 11–12

- Work Breakdown Structure and WBS Dictionary
- Network Diagram, PERT and the Critical Path

Labs 13–14

- Cost Estimating, Budget and Reserves
- Risk Register, RBS, EMV and Decision Tree

Requirements Elicitation, Mind Mapping and the Traceability Matrix

LAB 9

Turn the six charter requirements into a testable requirement set. Match elicitation techniques to stakeholder groups rather than defaulting to interviews, then expand them into a mind map. From the map write a 31-line register (REQ-001 to REQ-031) across the business, stakeholder, functional, non-functional, transition and quality categories. Build a bidirectional RTM from source to deliverable and test.

You'll build

A requirements mind map, a 31-requirement register with sourced, prioritised and testable entries, and a bidirectional RTM carrying the 13 compliance requirements to the governance gate - artifacts/09-requirements-mindmap.md, artifacts/09-requirements-register.md and artifacts/09-rtm.md.

Tools: Mind Mapping, Design Thinking (optional)

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 1 OF 7



1

Choose elicitation techniques deliberately, matching each technique to the stakeholder group and noting its weakness

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 2 OF 7

A green circle with a white number 2 inside, indicating the second step of the process.

Build the requirements mind map in the live Mind Mapping tool from the six charter requirements

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 3 OF 7



3

Write the requirements register across all six categories, with a source, priority and acceptance criterion per line

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 4 OF 7



4

Set requirement attributes and versioning, and define the route by which a baselined requirement changes

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 5 OF 7



5

Build the bidirectional requirements traceability matrix from source through deliverable to test

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 6 OF 7



6

Work through what bidirectional traceability buys you when a requirement is challenged or dropped

Requirements Elicitation, Mind Mapping and the Traceability Matrix

STEP 7 OF 7



7

Answer the exam-style scenarios on elicitation choice, requirement quality and traceability

Requirements Elicitation, Mind Mapping and the Traceability Matrix

Test it

Every requirement in the register has a unique ID, a named source, a MoSCoW priority and an acceptance criterion stated as a number or a pass/fail condition; no non-functional requirement is left unmeasurable. Each of the 13 compliance requirements C-01 to C-13 can be traced forward from the RTM to a deliverable and a test, and every requirement traces backward to a charter item or stakeholder.

MoSCoW, Kano and the Product Backlog

LAB 10

Prioritise the Lab 09 requirements with two models that deliberately disagree. Apply MoSCoW under discipline rules - legally required, contractually required, or non-viable without it - to separate Musts from wishes. Classify the same features on Kano as basic, performance, excitement, indifferent or reverse, and resolve each conflict. Write a story backlog in points and a capacity-based release plan.

You'll build

A MoSCoW classification, a Kano classification of 14 features, a 21-plus story backlog with points and traces to REQ IDs, and a capacity-based release plan - artifacts/10-moscow.md, artifacts/10-kano.md, artifacts/10-product-backlog.md and artifacts/10-release-plan.md.

Tools: Scrum Simulator

MoSCoW, Kano and the Product Backlog

STEP 1 OF 8

1

Apply MoSCoW to the Lab 09 requirements using the legal, contractual and viability discipline tests

MoSCoW, Kano and the Product Backlog

STEP 2 OF 8



2

Classify the features on Kano as basic, performance, excitement, indifferent or reverse

MoSCoW, Kano and the Product Backlog

STEP 3 OF 8



3

Examine where MoSCoW and Kano disagree, and resolve each conflict with a stated rule

MoSCoW, Kano and the Product Backlog

STEP 4 OF 8



4

Write the product backlog as user stories with acceptance criteria and traces to requirement IDs

MoSCoW, Kano and the Product Backlog

STEP 5 OF 8



5

Estimate with story points rather than hours, and justify relative sizing

MoSCoW, Kano and the Product Backlog

STEP 6 OF 8



6

Build the release plan against real team capacity and verify the Must-haves land before the gate

MoSCoW, Kano and the Product Backlog

STEP 7 OF 8



Make the AI recommender trade-off numeric using story points against benefit value

MoSCoW, Kano and the Product Backlog

STEP 8 OF 8



8

Answer the exam-style scenarios on prioritisation, value delivery and backlog management

MoSCoW, Kano and the Product Backlog

Test it

Every Must in the MoSCoW list survives the three discipline tests, and no more than the agreed share of total story points sits in Must. The release plan arithmetic reconciles: summed Must-have points divided by measured team velocity fits inside the sprints available before the gate, with the shortfall or slack stated explicitly. Each Kano basic is also a MoSCoW Must.

Work Breakdown Structure and WBS Dictionary

LAB 11

Decompose the Contoso scope into a deliverable-oriented WBS to three levels under the 100% rule, using the 8/80 rule and the estimate-and-assign test to decide when to stop. Every charter in-scope item must appear in the tree, no out-of-scope item anywhere. Write dictionary entries carrying effort in person-days, cost, predecessors, acceptance criteria and a control account, then assemble the scope baseline.

You'll build

A three-level deliverable-oriented WBS, a WBS dictionary with costed and estimated work-package entries, and the assembled scope baseline - artifacts/11-wbs.md, artifacts/11-wbs-dictionary.md and artifacts/11-scope-baseline.md.

Tools: WBS decomposition, 8/80 rule, Rolling wave planning

Work Breakdown Structure and WBS Dictionary

STEP 1 OF 8



1

Confirm the inputs and the scope boundary from the charter, register and backlog

Work Breakdown Structure and WBS Dictionary

STEP 2 OF 8

2

Build the WBS to three levels, deliverable-oriented, under the 100% rule

Work Breakdown Structure and WBS Dictionary

STEP 3 OF 8



3

Trace the compliance workstream back to the environmental factor that created it

Work Breakdown Structure and WBS Dictionary

STEP 4 OF 8



4

Write the WBS dictionary entries with effort, cost, predecessors and acceptance criteria

Work Breakdown Structure and WBS Dictionary

STEP 5 OF 8



5

Place control accounts and planning packages for the work not yet knowable

Work Breakdown Structure and WBS Dictionary

STEP 6 OF 8



6

Assemble the scope baseline and state exactly what it contains

Work Breakdown Structure and WBS Dictionary

STEP 7 OF 8



Reconcile the WBS with the product backlog so the two do not duplicate each other

Work Breakdown Structure and WBS Dictionary

STEP 8 OF 8



8

Answer the exam-style scenarios on decomposition, the 100% rule and baseline control

Work Breakdown Structure and WBS Dictionary

Test it

Roll the work-package effort and cost estimates up each branch and confirm they sum to the level-2 and level-1 totals with no orphan work and no double counting - this is the 100% rule proved arithmetically. Check every charter in-scope item appears exactly once in the tree, that no out-of-scope item appears at all, and that each work package sits inside the 8/80 hour band and can be assigned to one owner.

Network Diagram, PERT and the Critical Path

LAB 12

Convert the Lab 11 work packages into an activity list, sequence them with the four dependency types, and draw the precedence network. Compute the forward and backward pass by hand for early/late start and finish, total and free float, taking the critical path as the zero-float chain. Apply three-point estimating with the triangular and PERT beta formulas, then compare crashing against fast-tracking on cost per day.

You'll build

An activity list, a precedence network, a critical path with float per activity, PERT durations with variance, a costed compression analysis and the schedule baseline - [artifacts/12-activity-list.md](#), [artifacts/12-network-diagram.md](#), [artifacts/12-critical-path.md](#), [artifacts/12-schedule-baseline.md](#).

Tools: PERT, Critical Path Method, Forward and backward pass, Crashing and fast-tracking

Network Diagram, PERT and the Critical Path

STEP 1 OF 9



1

Derive the activity list from the WBS work packages, converting deliverables into activities

Network Diagram, PERT and the Critical Path

STEP 2 OF 9

A green circle with a white number 2 inside, indicating the second step of the process.

Understand the dependencies just declared, separating mandatory from discretionary logic

Network Diagram, PERT and the Critical Path

STEP 3 OF 9



3

Draw the precedence network diagram with the four dependency types

Network Diagram, PERT and the Critical Path

STEP 4 OF 9



4

Compute the forward and backward pass, then total float and free float per activity

Network Diagram, PERT and the Critical Path

STEP 5 OF 9



5

Apply three-point estimating with both formulas and compute the project confidence range from variance

Network Diagram, PERT and the Critical Path

STEP 6 OF 9



6

Compress the schedule by comparing crashing and fast-tracking on cost per day saved

Network Diagram, PERT and the Critical Path

STEP 7 OF 9



7

Apply the lessons-learned constraint that compliance review precedes UAT, and price the parallel alternative

Network Diagram, PERT and the Critical Path

STEP 8 OF 9



8

Build the schedule baseline from the accepted network and durations

Network Diagram, PERT and the Critical Path

STEP 9 OF 9

9

Answer the exam-style scenarios on float, critical path and compression

Network Diagram, PERT and the Critical Path

Test it

The backward pass must reproduce the forward pass: project late finish equals early finish, and every critical-path activity shows zero total float. Confirm the PERT roll-up - expected duration near 123 days, standard deviation about 4.73 days, a 95.45% range of roughly 113.5 to 132.5 days - by summing critical-path variances. The compression must shorten the critical path, not a float-bearing chain.

Cost Estimating, Budget and Reserves

LAB 13

Select estimating techniques against the information available and state each one's accuracy range, then build a bottom-up estimate from the Lab 11 work packages and reconcile it to the charter budget. Separate contingency from management reserve - who authorises release, which sits in the cost baseline - and size contingency by expected monetary value. Time-phase into a cost baseline S-curve.

You'll build

A bottom-up cost estimate reconciled to the charter, an EMV-sized contingency reserve, a time-phased cost baseline S-curve with the funding requirement above it, and a cost-of-quality analysis - artifacts/13-cost-estimates.md, artifacts/13-cost-baseline.md and artifacts/13-reserve-analysis.md.

Tools: Statistics, Bottom-up estimating, Expected Monetary Value

Cost Estimating, Budget and Reserves

STEP 1 OF 7



1

Build the bottom-up estimate from the WBS work packages and reconcile it to the charter budget

Cost Estimating, Budget and Reserves

STEP 2 OF 7

A green circle with a white number 2 inside, indicating the second step of the process.

Distinguish contingency reserve from management reserve, including authority and baseline placement

Cost Estimating, Budget and Reserves

STEP 3 OF 7



3

Size the contingency reserve by expected monetary value rather than by percentage

Cost Estimating, Budget and Reserves

STEP 4 OF 7



4

Build the time-phased cost baseline as an S-curve

Cost Estimating, Budget and Reserves

STEP 5 OF 7



5

Perform funding limit reconciliation against the cash-flow profile

Cost Estimating, Budget and Reserves

STEP 6 OF 7



6

Analyse the cost of quality across prevention, appraisal, internal failure and external failure

Cost Estimating, Budget and Reserves

STEP 7 OF 7

7

Answer the exam-style scenarios on estimating accuracy, reserves and baseline versus budget

Cost Estimating, Budget and Reserves

Test it

The money must add up both ways: work-package estimates plus contingency equals the cost baseline, and cost baseline plus management reserve equals the total budget. Confirm the EMV sizing by recomputing probability times impact per priced risk. The cumulative S-curve must end at the cost baseline total, and funding limit reconciliation must show no period where planned spend exceeds the authorised tranche.

Risk Register, RBS, EMV and Decision Tree

LAB 14

Build a risk breakdown structure and use it to identify risks systematically, tracing them from the Lab 02 environment scan and the Lab 06 assumption log - every assumption is a latent risk. Write each in cause-event-effect form, then assess the register on a probability/impact matrix and by expected monetary value. Give each a response from the ten strategies and solve a decision tree on EMV.

You'll build

A risk breakdown structure, a risk register with cause-event-effect statements and owners, a plotted probability/impact matrix, a full EMV analysis and a solved decision tree - artifacts/14-risk-register.md, artifacts/14-rbs.md, artifacts/14-pi-matrix.md and artifacts/14-emv-analysis.md.

Tools: System Thinking, EMV, Decision Tree, Probability/Impact Matrix

Risk Register, RBS, EMV and Decision Tree

STEP 1 OF 10



Build the risk breakdown structure and use it to drive systematic identification

Risk Register, RBS, EMV and Decision Tree

STEP 2 OF 10

A green circle with a white number 2 inside, indicating the second step of the process.

Trace risks from the Lab 02 PESTLE/TECOP factors and the Lab 06 assumption log

Risk Register, RBS, EMV and Decision Tree

STEP 3 OF 10



3

Write risk statements in cause-event-effect form

Risk Register, RBS, EMV and Decision Tree

STEP 4 OF 10

4

Build the risk register with probability, impact, owner and trigger per risk

Risk Register, RBS, EMV and Decision Tree

STEP 5 OF 10



5

Plot the probability/impact matrix and rank the risks by score

Risk Register, RBS, EMV and Decision Tree

STEP 6 OF 10



6

Apply all ten response strategies - five for threats and five for opportunities

Risk Register, RBS, EMV and Decision Tree

STEP 7 OF 10

7

Compute expected monetary value for each priced risk and for the portfolio

Risk Register, RBS, EMV and Decision Tree

STEP 8 OF 10



8

Build and solve a decision tree for a real Contoso decision and defend the recommendation

Risk Register, RBS, EMV and Decision Tree

STEP 9 OF 10

9

Distinguish qualitative from quantitative analysis, and a risk from an issue

Risk Register, RBS, EMV and Decision Tree

STEP 10 OF 10

10

Answer the exam-style scenarios on response strategy selection, EMV and secondary risk

Risk Register, RBS, EMV and Decision Tree

Test it

Recompute each risk's EMV as probability times impact, summing threats and opportunities separately - the net portfolio EMV must reconcile with the contingency reserve sized in Lab 13. Solve the decision tree by rolling values back from the leaves; the recommended branch must have the highest expected value net of cost. Every register row carries an owner, a trigger and one of the ten strategies.

Recap — Plan the Project

1

You can now: Process T2 - Develop and manage project scope; Process T1 - Develop an integrated project management plan.

2

You can now: Process T3 - Help ensure value-based delivery; Process T2 - Develop and manage project scope.

3

You can now: Process T2 - Develop and manage project scope (break down scope); Process T1 - Develop an integrated project management plan.

4

You can now: Process T8 - Plan and manage schedule.

5

You can now: Process T6 - Plan and manage finance; Process T1 - Develop an integrated project management plan.

6

You can now: Business Environment T5 - Plan and manage risk; Business Environment T4 - Support organisational change.

TOPIC 04

Lead the Project Team

Leadership · motivation · conflict · coaching · communication

Key Concepts — Lead the Project Team

1

Tuckman's forming, storming, norming, performing and adjourning gives you a diagnosis, and each stage calls for a different leadership response.

2

Situational leadership, servant leadership and motivation theory (Maslow, Herzberg, McClelland, Theory X/Y) drive team performance.

3

RACI removes ambiguity: exactly one Accountable per work package, Responsible parties who do the work.

4

The five conflict modes — collaborate, compromise, smooth, force, withdraw — are chosen by context; collaborate/problem-solve is preferred.

5

The channels formula $n(n-1)/2$ shows why communication complexity grows faster than team size.

6

A communication plan defines who needs what information, in what format, how often, and through which channel.

Hands-On Labs — Lead the Project Team

Labs 15–16

- RACI Responsibility Assignment Matrix and Resource Plan
- Tuckman Diagnosis, Motivation Theory and Leadership Style

Labs 17–18

- Conflict Resolution: Five Modes, Leas' Levels and Negotiation
- Communication Plan, Channels and Status Reporting

RACI Responsibility Assignment Matrix and Resource Plan

LAB 15

Build a RACI matrix from the Lab 11 WBS work packages, defending every assignment against the one-accountable rule. The supplied draft carries five deliberate defects - missing accountability, two accountables on a row, over-consultation, an unassigned role and an unowned work package - which you audit out and fix. Build the resource histogram and distinguish levelling from smoothing.

You'll build

A defect-free RACI matrix over the WBS work packages, a documented audit of the five structural defects with fixes, and a resource plan with a levelled histogram - artifacts/15-raci-matrix.md, artifacts/15-resource-plan.md and artifacts/15-raci-issues.md.

Tools: RACI Matrix

RACI Responsibility Assignment Matrix and Resource Plan

STEP 1 OF 9



1

Assemble the inputs from the WBS dictionary, team charter and stakeholder register

RACI Responsibility Assignment Matrix and Resource Plan

STEP 2 OF 9



2

Draft the matrix, accepting that the supplied draft contains deliberate defects

RACI Responsibility Assignment Matrix and Resource Plan

STEP 3 OF 9

3

Enter the matrix in the live RACI Matrix tool

RACI Responsibility Assignment Matrix and Resource Plan

STEP 4 OF 9



4

Audit the matrix and find the five structural defects

RACI Responsibility Assignment Matrix and Resource Plan

STEP 5 OF 9



5

Compare before and after in numbers, counting assignments per row and per role

RACI Responsibility Assignment Matrix and Resource Plan

STEP 6 OF 9



6

Build the resource plan and the resource histogram

RACI Responsibility Assignment Matrix and Resource Plan

STEP 7 OF 9



Level or smooth the over-allocation, and state which technique can move the critical path

RACI Responsibility Assignment Matrix and Resource Plan

STEP 8 OF 9



8

Connect the RACI to the communication plan so the C and I entries drive the distribution list

RACI Responsibility Assignment Matrix and Resource Plan

STEP 9 OF 9

9

Answer the exam-style scenarios on accountability, over-consultation and levelling versus smoothing

RACI Responsibility Assignment Matrix and Resource Plan

Test it

Audit the corrected matrix row by row and column by column: every work-package row shows exactly one A and at least one R, no row is empty, and no role column is empty. Compare the histogram before and after: levelling must remove every allocation above 100%, and the learner must state whether the end date moved (levelling) or held (smoothing).

Tuckman Diagnosis, Motivation Theory and Leadership Style

LAB 16

Diagnose the team's Tuckman stage from the sprint 3 behavioural evidence rather than elapsed time, naming structural causes not personalities, then select the leader actions that move a team out of storming. Apply six motivation theories - Maslow, Herzberg, McGregor, McClelland, Vroom and Self-Determination - for one action each. Compute a Vroom score as expectancy times instrumentality times valence.

You'll build

A Tuckman diagnosis with structural causes and leader actions, per-member motivation profiles with one action from each of the six theories, and a leadership style map - artifacts/16-tuckman-diagnosis.md, artifacts/16-motivation-plan.md and artifacts/16-leadership-style-plan.md.

Tools: Tuckman model, Situational Leadership, Vroom Expectancy calculation

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 1 OF 8



1

Read the sprint 3 evidence and record the observed behaviours without interpreting them yet

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 2 OF 8



2

Diagnose the Tuckman stage and name the structural causes behind it

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 3 OF 8



3

Choose the leader actions that move a team out of storming

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 4 OF 8

4

Apply the six motivation theories, producing one specific action from each

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 5 OF 8



5

Build per-member motivation profiles, computing the Vroom expectancy score numerically

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 6 OF 8



6

Map a situational leadership style to each member's competence and commitment

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 7 OF 8

7

Distinguish servant leadership from laissez-faire behaviourally and by outcome

Tuckman Diagnosis, Motivation Theory and Leadership Style

STEP 8 OF 8



8

Answer the exam-style scenarios on stage diagnosis, Herzberg and style selection

Tuckman Diagnosis, Motivation Theory and Leadership Style

Test it

The diagnosis cites at least three observed sprint 3 behaviours and does not rest on how long the team has existed. Each Vroom score is computed as expectancy times instrumentality times valence, with the lowest factor identified as the one to fix - a low score with high valence and instrumentality points at expectancy - and the action must target that factor. Every member carries a different style justification.

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

LAB 17

Place the five conflict-handling modes on the assertiveness and cooperativeness axes, state each one's legitimate use, and identify the cases where collaborate - usually the keyed answer - is wrong. Diagnose intensity on Leas' five levels from the language people use. Run four Contoso role-plays, recording each in a conflict log. Close on negotiation: positions versus interests, BATNA and ZOPA.

You'll build

A conflict log recording level, mode, intervention and outcome per incident, observer notes and debriefs for the four role-plays, and a resolution plan with a BATNA and ZOPA analysis - artifacts/17-conflict-log.md, artifacts/17-roleplay-notes.md and artifacts/17-resolution-plan.md.

Tools: Thomas-Kilmann five modes, Leas' Levels, BATNA and ZOPA analysis

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 1 OF 8



1

Build the conflict log with columns for level, mode, intervention and outcome

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 2 OF 8

2

Run role-play 1: two developers on architecture at Leas' level 1 to 2

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 3 OF 8



3

Run role-play 2: Head of Sales versus product owner on scope at level 3

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 4 OF 8



4

Run role-play 3: QA lead versus dev lead on the definition of done at level 2 to 3

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 5 OF 8



5

Run role-play 4: admin staff versus Ops Manager on redundancy, escalating and high emotion

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 6 OF 8



6

Build the resolution plan from the four logged conflicts

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 7 OF 8

7

Apply negotiation: separate positions from interests, and compute BATNA and ZOPA

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

STEP 8 OF 8



8

Answer the exam-style scenarios on mode selection and escalation thresholds

Conflict Resolution: Five Modes, Leas' Levels and Negotiation

Test it

Every conflict log row names a Leas' level, the mode chosen, and why that mode beat collaborate - a level 4 or 5 entry that still selects collaborate is a failed diagnosis. The negotiation analysis states each party's BATNA explicitly and shows the ZOPA as a numeric range between the two reservation points, or states that no ZOPA exists and no deal is available.

Communication Plan, Channels and Status Reporting

LAB 18

Compute the communication channels with $N(N-1)/2$ and derive the implication of its superlinear growth. Select push, pull or interactive per stakeholder, place communications in the formal/informal by written/verbal grid, and build a plan where every row carries a feedback loop. Write the week-14 status three ways for three audiences, noticing that a CPI of 0.919 breaches the governance threshold.

You'll build

A communication plan with computed channel counts, a push/pull/interactive assignment and a named feedback loop per stakeholder row, plus three audience-specific week-14 status reports written from one set of numbers - artifacts/18-communication-plan.md and artifacts/18-status-reports.md.

Tools: $N(N-1)/2$ channels formula, Sender-receiver model, Status reporting

Communication Plan, Channels and Status Reporting

STEP 1 OF 9



1

Compute the communication channels with $N(N-1)/2$ and derive the growth implication

Communication Plan, Channels and Status Reporting

STEP 2 OF 9



2

Choose push, pull or interactive for each stakeholder and justify the choice

Communication Plan, Channels and Status Reporting

STEP 3 OF 9



3

Place communications in the formal/informal by written/verbal grid

Communication Plan, Channels and Status Reporting

STEP 4 OF 9



4

Build the communication plan with a named feedback loop on every row

Communication Plan, Channels and Status Reporting

STEP 5 OF 9



5

Trace a message through the sender-receiver model and name the real noise sources

Communication Plan, Channels and Status Reporting

STEP 6 OF 9



6

Work through the week-14 status data and compute the performance indices

Communication Plan, Channels and Status Reporting

STEP 7 OF 9



7

Write the same status three ways for three different audiences from one set of numbers

Communication Plan, Channels and Status Reporting

STEP 8 OF 9



8

Compare a bad status report with a good one and name what makes the difference

Communication Plan, Channels and Status Reporting

STEP 9 OF 9



9

Answer the exam-style scenarios on channels, communication method and escalation

Communication Plan, Channels and Status Reporting

Test it

Recompute $N(N-1)/2$ before and after the proposed additions and confirm the channel count grows superlinearly - state the delta, not just the totals. Every plan row names a sender, audience, method, frequency and feedback loop. The three status reports reconcile to identical figures despite differing in emphasis, and the CPI of 0.919 is flagged against the governance cost variance threshold with the escalation named.

Recap — Lead the Project Team

1

You can now: People T3 - Lead the project team (establish clear roles and responsibilities within the team); Process T4 - Plan and manage resources.

2

You can now: People T3 - Lead the project team; People T1 - Develop a common vision.

3

You can now: People T2 - Manage conflicts.

4

You can now: People T8 - Plan and manage communication; People T4 - Engage stakeholders.

TOPIC 05

Monitor and Control the Project

Earned value · flow metrics · quality control · change control · issues

Key Concepts — Monitor and Control the Project

1

Earned value management combines scope, schedule and cost into CPI, SPI, EAC and VAC — it tells you THAT something is wrong, not why.

2

Schedule compression is a choice between crashing (adds cost) and fast-tracking (adds risk); both must be justified against the baseline.

3

Kanban flow metrics — WIP limits, lead time and cycle time — expose bottlenecks that a Gantt chart hides.

4

Root cause analysis with 5 Whys, fishbone and Pareto finds the vital few causes behind most of the pain.

5

Statistical process control separates assignable causes from common-cause noise, and a stable process can still fail its specification.

6

Integrated change control evaluates every change against all baselines before approval — the change control board decides, the PM does not.

Hands-On Labs — Monitor and Control the Project

Labs 19–20

- Kanban Board with WIP Limits and Lead/Cycle Time
- Earned Value Management and Schedule Compression

Labs 21–22

- Root Cause Analysis with 5 Whys, Fishbone and Pareto
- SPC Control Chart and Statistical Process Analysis

Kanban Board with WIP Limits and Lead/Cycle Time

LAB 19

Build a nine-column Kanban board mirroring Contoso's workflow, separating active from queue columns so waiting time becomes visible. Set a justified WIP limit per column, load 18 story cards with backlog-entry, work-start and acceptance dates, and compute lead time, cycle time, throughput and flow efficiency. Apply Little's Law, then find the bottleneck and apply the five Theory of Constraints steps.

You'll build

artifacts/19-kanban-board.md with the nine-column workflow and WIP limits; **artifacts/19-flow-metrics.md** with lead and cycle time, throughput, flow efficiency and the Little's Law verification; **artifacts/19-bottleneck-analysis.md** with the utilisation table and Theory of Constraints response.

Tools: Kanban Board

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 1 OF 8



Design the workflow columns, separating active stages from queue stages

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 2 OF 8



2

Set and justify a WIP limit for every column against the team composition

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 3 OF 8



3

Load the 18 real cards with backlog-entry, work-start and acceptance dates

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 4 OF 8



4

Compute the flow metrics - lead time, cycle time, throughput and flow efficiency

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 5 OF 8



5

Apply Little's Law to verify stability and forecast the remaining release

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 6 OF 8



6

Find the bottleneck from column utilisation and apply the Theory of Constraints steps

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 7 OF 8



Read the cumulative flow diagram and interpret band width, slope and convergence

Kanban Board with WIP Limits and Lead/Cycle Time

STEP 8 OF 8



8

Answer the exam-style scenarios on Little's Law, WIP limits and lead versus cycle time

Kanban Board with WIP Limits and Lead/Cycle Time

Test it

Mean lead time comes to 13.11 days against a mean cycle time of 6.94 days, a flow efficiency of 52.9%; Little's Law implies a WIP of about 6.25 against an observed 6 on day 12; Test is the bottleneck on its 100% utilisation, the 97% queue in front of it and the collapse to 33% downstream; and the learner rejects raising the In Test WIP limit, explaining why it would lengthen cycle time.

Earned Value Management and Schedule Compression

LAB 20

From the SGD 480,000 Contoso cost baseline, validate that cumulative planned value at the final period equals BAC, then compute CV, SV, CPI and SPI at the week-14 status date. Calculate all three EAC variants, choose the defensible one on evidence, and derive ETC, VAC and both TCPI figures. Analyse the seven-period CPI/SPI trend and choose between crashing and fast-tracking from the crash-cost table.

You'll build

artifacts/20-evm-calculations.md with CV, SV, CPI and SPI plus the seven-period trend table; artifacts/20-forecast.md with all three EAC variants, ETC, VAC and both TCPI figures; artifacts/20-compression-decision.md with the five-day crash recovery, its cost and the rejection of fast-tracking.

Tools: Statistics

Earned Value Management and Schedule Compression

STEP 1 OF 8



1

Take the cost baseline and actuals, and validate that cumulative PV equals BAC

Earned Value Management and Schedule Compression

STEP 2 OF 8

2

Compute the variances and indices at week 14 - CV, SV, CPI and SPI

Earned Value Management and Schedule Compression

STEP 3 OF 8



3

Forecast the outcome with all three EAC variants, plus ETC, VAC and TCPI

Earned Value Management and Schedule Compression

STEP 4 OF 8



4

Analyse the seven-period CPI/SPI trend rather than the single snapshot

Earned Value Management and Schedule Compression

STEP 5 OF 8



5

Check the result against the escalation thresholds and decide whether to escalate

Earned Value Management and Schedule Compression

STEP 6 OF 8



6

Decide between crashing and fast-tracking using the crash-cost table

Earned Value Management and Schedule Compression

STEP 7 OF 8

7

Write the variance report in sponsor format with three costed options and a recommendation

Earned Value Management and Schedule Compression

STEP 8 OF 8



8

Answer the exam-style scenarios on EAC selection, EVM reading and trend interpretation

Earned Value Management and Schedule Compression

Test it

Cumulative PV at period 12 equals SGD 480,000; CV = -22,000 and SV = -19,000 are in dollars not days; CPI = 0.919 and SPI = 0.929, CPI carried to four decimals into EAC = SGD 522,420 and VAC = -SGD 42,420; the TCPI of 1.105 against CPI 0.919 makes recovery not credible; the crash selects J and O for SGD 15,500, rejecting non-critical F; and escalation is raised on the forecast, not the actual variance.

Root Cause Analysis with 5 Whys, Fishbone and Pareto

LAB 21

Take the defect log for sprints 1 to 7 - the period over which CPI declined monotonically - and run the three root cause tools in sequence. Pareto on defect frequency isolates the vital few inside the 80% cut; a second Pareto by rework hours reveals where frequency and cost disagree. A fishbone generates causes across the six categories, and 5 Whys drives from symptom to root cause, verified with the removal test.

You'll build

artifacts/21-pareto-analysis.md with the frequency and cost rankings and cumulative percentages;
artifacts/21-fishbone.md with two causes per category; **artifacts/21-5whys.md** with the causal chain;
artifacts/21-corrective-actions.md separating corrective from preventive action.

Tools: Pareto Chart, Fishbone Diagram, 5 Whys

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 1 OF 8



Take the defect data covering sprints 1 to 7 with counts, rework hours and cost

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 2 OF 8



2

Build the Pareto chart on defect frequency and locate the 80% cut

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 3 OF 8

3

Re-run Pareto by rework cost and notice where the two rankings disagree

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 4 OF 8



4

Fishbone the top category across all six cause categories

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 5 OF 8



5

Drive from symptom to root cause with 5 Whys

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 6 OF 8



6

Verify the root cause with the therefore test and the removal test before acting

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 7 OF 8



Design corrective, preventive and defect-repair actions and assign owners

Root Cause Analysis with 5 Whys, Fishbone and Pareto

STEP 8 OF 8



8

Answer the exam-style scenarios on root cause versus symptom and action type

Root Cause Analysis with 5 Whys, Fishbone and Pareto

Test it

Cumulative percentages reach 100% with the 80% cut at rank 5 on 81.55%; both Pareto runs rank acceptance criteria and browser compatibility first and second; every fishbone category carries two causes; the root cause is a system or process failure rather than an individual's shortcoming and passes the removal test; and the causal chain connects it back to the CPI decline found by earned value analysis.

SPC Control Chart and Statistical Process Analysis

LAB 22

Plot 20 daily measurements of median registration completion time and compute the centre line and control limits from the data itself, using the moving-range method rather than the sample standard deviation. Apply the out-of-control rules including the rule of seven and identify where the process shifts. Separate common from assignable cause, then assess capability - the process is in control yet fails specification.

You'll build

artifacts/22-control-chart.md with the individuals chart, centre line and control limits; **artifacts/22-stability-verdict.md** with the rule-by-rule out-of-control analysis and the named assignable cause; **artifacts/22-capability-summary.md** with the capability indices against the charter target.

Tools: SPC / Control Chart, Statistics

SPC Control Chart and Statistical Process Analysis

STEP 1 OF 8



Take the 20 daily measurements of median registration completion time

SPC Control Chart and Statistical Process Analysis

STEP 2 OF 8



2

Compute the centre line and control limits from the moving ranges

SPC Control Chart and Statistical Process Analysis

STEP 3 OF 8



3

Plot the individuals chart against the computed limits

SPC Control Chart and Statistical Process Analysis

STEP 4 OF 8



4

Apply the out-of-control rules, including the rule of seven, run by run

SPC Control Chart and Statistical Process Analysis

STEP 5 OF 8



5

Separate common cause from assignable cause and decide the response to each

SPC Control Chart and Statistical Process Analysis

STEP 6 OF 8



6

Assess capability against the charter success criterion and distinguish control from specification limits

SPC Control Chart and Statistical Process Analysis

STEP 7 OF 8



7

Write the quality verdict and the corrective actions

SPC Control Chart and Statistical Process Analysis

STEP 8 OF 8



8

Answer the exam-style scenarios on tampering, signals and capability

SPC Control Chart and Statistical Process Analysis

Test it

The centre line comes to 4.053 with MR-bar 0.3286, sigma 0.2913 and limits of 3.179 and 4.927, computed from the stable period not all 20 points; sigma is MR-bar divided by 1.128; the rule of seven is checked on days 1 to 15, longest run 2; days 16 to 20 are assignable cause with a dated explanation matching the shift; and the negative Cpk leads to re-centring rather than tightening.

Recap — Monitor and Control the Project

1

You can now: Process T9 - Evaluate project status; Business Environment T4 - Remove impediments and manage issues.

2

You can now: Process T9 - Evaluate project status; Process T6 - Plan and manage budget and resources.

3

You can now: Business Environment T4 - Remove impediments and manage issues; Process T7 - Plan and manage quality of products and deliverables; Business Environment T6 - Support continuous improvement.

4

You can now: Process T7 - Plan and manage quality of products and deliverables; Process T9 - Evaluate project status.

TOPIC 06

Close the Project

Acceptance · transition · benefits realisation · lessons learned · exam strategy

Key Concepts — Close the Project

1

Closure requires formal stakeholder acceptance against documented criteria, not merely the end of the work.

2

Transition readiness moves the deliverable to operations with the support, documentation and training that make it sustainable.

3

Benefits realisation is measured after go-live against the business case — the project ends, the benefits do not.

4

Lessons learned only count when they update an organisational process asset that the next project inherits.

5

Knowledge transfer protects the organisation from the loss of the team that built the thing.

6

Exam strategy: read the last sentence first, identify the development approach in play, and eliminate answers that bypass the process.

Hands-On Labs — Close the Project

Lab 23

- Project Closure, Benefits Realisation and Lessons Learned

Lab 24

- Capstone - Integrated Project Management Plan and Mock Exam

Project Closure, Benefits Realisation and Lessons Learned

LAB 23

Close the Contoso project against the four closure enablers. Define closure criteria first, run Control Quality against specification before seeking acceptance through Validate Scope, then validate operational readiness. Write the final report against all eight charter success criteria - including the one missed - and reconcile the final cost. Hand every benefit to a named owner with a post-closure review date.

You'll build

artifacts/23-closure-checklist.md separating procurement from administrative closure; **artifacts/23-final-report.md** with the eight criteria and cost reconciliation; **artifacts/23-lessons-learned.md**; **artifacts/23-benefits-realisation.md** handing B1 to B4 to named owners; **artifacts/23-transition-plan.md**.

Tools: Closure checklist, Benefits realisation register, Lessons learned register

Project Closure, Benefits Realisation and Lessons Learned

STEP 1 OF 9



1

Define the closure criteria before closing anything

Project Closure, Benefits Realisation and Lessons Learned

STEP 2 OF 9



2

Verify scope with Control Quality before seeking acceptance

Project Closure, Benefits Realisation and Lessons Learned

STEP 3 OF 9



3

Obtain formal acceptance from the customer through Validate Scope

Project Closure, Benefits Realisation and Lessons Learned

STEP 4 OF 9



4

Validate readiness for transition to operations

Project Closure, Benefits Realisation and Lessons Learned

STEP 5 OF 9



5

Write the final report against all eight charter success criteria

Project Closure, Benefits Realisation and Lessons Learned

STEP 6 OF 9



6

Hand over the benefits to named owners with review dates beyond project end

Project Closure, Benefits Realisation and Lessons Learned

STEP 7 OF 9



Capture lessons learned that each update a named organisational process asset

Project Closure, Benefits Realisation and Lessons Learned

STEP 8 OF 9



Plan the knowledge transfer and release the team

Project Closure, Benefits Realisation and Lessons Learned

STEP 9 OF 9

A green circle with a white number 9 inside, indicating the step number.

Answer the exam-style scenarios on closure sequence, acceptance and benefits ownership

Project Closure, Benefits Realisation and Lessons Learned

Test it

Control Quality verifies against specification first, Validate Scope obtains acceptance second; the checklist separates procurement from administrative closure; SC-2 is reported NOT MET with its consequence quantified as a SGD 42,529 annual benefit shortfall; the final cost reconciles to SGD 478,900 with contingency closing at zero exposure; and every benefit B1 to B4 has a named owner who is not the project manager.

Capstone - Integrated Project Management Plan and Mock Exam

LAB 24

Part A assembles twenty-three labs of artifacts into one integrated project management plan: a component index mapping all 25 components to the lab that produced each, the three baselines and the PMB accounted separately, and a twenty-assertion consistency audit. Part B is a 30-question mock practice exam in ECO 2026 format under a 40-minute timer, scored by domain against the published weights.

You'll build

artifacts/24-project-management-plan.md with the component index and twenty-assertion audit; **artifacts/24-mock-exam-answers.md** with all 30 answers recorded before the key was read; **artifacts/24-score-analysis.md**; **artifacts/24-exam-readiness-plan.md** with a dated plan for every domain below 70%.

Tools: Project management plan index, Consistency audit, Mock practice exam, Score analysis

Capstone - Integrated Project Management Plan and Mock Exam

STEP 1 OF 9



1

Build the plan index mapping all 25 components to the lab and artifact that produced each

Capstone - Integrated Project Management Plan and Mock Exam

STEP 2 OF 9



2

Account for the three baselines and the performance measurement baseline separately

Capstone - Integrated Project Management Plan and Mock Exam

STEP 3 OF 9



3

Run the twenty-assertion consistency audit and record every discrepancy found

Capstone - Integrated Project Management Plan and Mock Exam

STEP 4 OF 9



4

Prepare the five-minute executive presentation outline with timings per slide

Capstone - Integrated Project Management Plan and Mock Exam

STEP 5 OF 9



5

Sit the 30-question mock practice exam in ECO 2026 format under a 40-minute timer

Capstone - Integrated Project Management Plan and Mock Exam

STEP 6 OF 9



6

Score the mock practice exam by domain against the 33/41/26 weights

Capstone - Integrated Project Management Plan and Mock Exam

STEP 7 OF 9



7

Separate confident correct answers from flagged guesses and identify the weakest domain

Capstone - Integrated Project Management Plan and Mock Exam

STEP 8 OF 9



8

Write the remediation plan naming specific labs, ECO tasks, a dated action and a re-test date

Capstone - Integrated Project Management Plan and Mock Exam

STEP 9 OF 9



9

Complete the exam-day readiness checklist covering eligibility, pacing and answer selection

Capstone - Integrated Project Management Plan and Mock Exam

Test it

The plan index accounts for the three baselines plus the PMB, and the learner explains why management reserve requires a baseline change while contingency does not; the audit PV sum comes to SGD 480,000 with at least one genuine discrepancy recorded; the mock exam was taken in 40 minutes without looking at the key; and per-domain scores are compared against 33%, 41% and 26%.

Recap — Close the Project

1

You can now: Process T10 - Manage project closure and transitions; Business Environment T6 - Support organizational change and continuous improvement; People T7 - Help ensure knowledge transfer.

2

You can now: All domains - People, Process and Business Environment across the full ECO 2026 task list.



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Scope & Authorise

Scope project requirements and build a business case, charter and traceable scope baseline that authorise delivery.

2

Plan & Baseline

Develop an integrated project plan with realistic schedule, cost and resource baselines using PERT, critical path and reserve analysis.

3

Lead the Team

Lead a project team through its development stages, applying leadership style, motivation theory and structured conflict resolution.

4

Risk & Stakeholders

Analyse project risk quantitatively and engage stakeholders through a planned communication and escalation cadence.

5

Control & Close

Control delivery against baselines with earned value, flow metrics and statistical process control, then close the project and realise benefits.

6

Exam Readiness

Apply the PMI PMP Examination Content Outline (ECO 2026) across all three domains and sit a full-format mock exam with domain score analysis.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Your course material stays available at <https://lms-tms.tertiaryinfotech.com/>.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

THANK YOU FOR ATTENDING

Thank You!

You now have the practical skills from every lab in this course — keep applying them to your own work.